



Apartment Seed Starting Kit

Grow food in small spaces — from seed to harvest

\$9

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Apartment Seed Starting Kit

A Practical Guide for First-Time Indoor Growers

Seedwarden | seedwarden.com

You don't need a yard. You need a window, some containers, and a little patience. This guide gives you everything else.

Important Notice

This guide is educational content based on real growing experience. Your results will vary depending on your light conditions, indoor environment, and local climate. It is not professional agricultural advice. Use it as a starting point, expect a learning curve in your first season, and adjust based on what actually happens in your space.

How to Use This Guide

Read through it once before you buy anything. Then come back to the seed starting schedules when you're ready to plan. The troubleshooting section at the end is your first call when something goes sideways — and something usually does, especially in year one. That's normal. You're learning to read plants, and that takes a season or two.

Part 1: Seed Starting Schedules

The timing below is based on **weeks before your last frost date**. If you don't know yours, look up your zip code at the USDA Plant Hardiness Zone Map or type "[your city] last frost date" into any search engine. Write your last frost date here before you go further:

My last frost date: _____

Then count backwards on a calendar to find when to start each crop indoors.

The 12 Crops

1. Tomatoes

- Start indoors: 6–8 weeks before last frost
 - Transplant outdoors: After last frost, once nighttime temps stay above 50°F
 - Container size: 5-gallon minimum per plant; 10–15 gallon preferred for full-size varieties. Cherry tomatoes can work in 5-gallon.
 - Light: High light. South-facing window is borderline — you'll get better results with a grow light. Needs 8+ hours of direct or full-spectrum light daily.
 - Notes: Don't rush. Tomatoes started too early get leggy and rootbound before it's warm enough to transplant. 6 weeks is usually better than 8.
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2. Peppers (Sweet and Hot)

- Start indoors: 8–10 weeks before last frost
 - Transplant outdoors: 2 weeks after last frost (peppers are more cold-sensitive than tomatoes)
 - Container size: 3–5 gallon per plant
 - Light: High light. Same needs as tomatoes. A south-facing window can work in climates with strong spring sun; supplemental light improves yield.
 - Notes: Peppers are slow starters. Don't panic if germination takes 2–3 weeks. Bottom heat (a seedling heat mat) dramatically improves germination rates.
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3. Eggplant

- Start indoors: 8–10 weeks before last frost
 - Transplant outdoors: 2–3 weeks after last frost
 - Container size: 5-gallon minimum
 - Light: High light. Very similar to peppers; needs consistent warmth and strong light to thrive.
 - Notes: Eggplant is the most heat-loving of the common vegetables. In apartments with cool interior temperatures, a heat mat during germination is almost essential.
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4. Basil

- Start indoors: 4–6 weeks before last frost
 - Transplant outdoors: After last frost; basil is very cold-sensitive
 - Container size: 6–8" pot per plant (or several plants in a window box)
 - Light: High light. Basil grown in low light gets leggy, flavorless, and prone to fungal problems.
 - Notes: Start multiple cells — basil is easy to direct-sow but starting indoors lets you get a jump on the season. Pinch flower buds as soon as they appear to extend harvest.
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5. Lettuce

- Start indoors: 4–6 weeks before last frost (or direct sow as soon as you can work the container)
 - Transplant outdoors: 2–4 weeks before last frost (lettuce tolerates light frost)
 - Container size: 6–8" deep, any width. Window boxes work well.
 - Light: Medium light. One of the few vegetables that genuinely does well in a bright east- or west-facing window. Does not need direct sun all day.
 - Notes: Lettuce bolts (goes to seed) in heat. Plan for a spring and fall crop rather than trying to grow through summer. Succession-plant every 2–3 weeks for continuous harvest.
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6. Kale and Chard

- Start indoors: 4–6 weeks before last frost
 - Transplant outdoors: 3–4 weeks before last frost (cold-tolerant)
 - Container size: 8–12" deep per plant; 2–3 plants fit in a standard window box
 - Light: Medium to high. These are among the most forgiving crops for lower-light situations.
 - Notes: Both can be harvested as cut-and-come-again — take outer leaves, leave the center, and the plant keeps producing. A single kale plant can feed you for months.
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7. Cucumbers

- Start indoors: 3–4 weeks before last frost (no earlier — they grow fast and get rootbound quickly)
 - Transplant outdoors: After last frost; cucumbers don't tolerate cold
 - Container size: 5-gallon minimum. They need more root space than they look like they do.
 - Light: High light. Cucumbers need full sun and warmth to produce well.
 - Notes: Bush varieties (like Bush Pickle or Spacemaster) do better in containers than vining types. If you do grow vining cucumbers on a balcony, they'll need a trellis or cage.
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8. Beans (Bush)

- Start indoors: Direct sow only, 2 weeks after last frost — OR start 2–3 weeks before transplanting if you want a head start
 - Container size: 8–12" deep, 12"+ wide per plant
 - Light: High light
 - Notes: Beans resent root disturbance. If starting indoors, use biodegradable pots (newspaper pots, peat pots, or soil blocks) so you can transplant without disturbing roots. Bush beans don't need support and are better for containers than pole beans.
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9. Radishes

- Start indoors: Not recommended — direct sow in containers as soon as last frost has passed (or 4–6 weeks before in spring, since they're frost-tolerant)
 - Container size: 6" deep minimum; standard window box works fine
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- Light: Medium to high
- Notes: Radishes mature in 22–30 days. They're the best crop for impatient gardeners and great for filling gaps between slower-growing plants. Sow every 2 weeks for continuous harvest.

10. Herbs: Parsley and Cilantro

- Start indoors: 6–8 weeks before last frost (both are slow to germinate)
- Transplant outdoors: 2–3 weeks before last frost (both tolerate cool temps)
- Container size: 6–8" pot, or together in a window box
- Light: Medium. Both tolerate partial shade better than most vegetables.
- Notes: Soak parsley seeds overnight before planting to speed germination — they can take 3 weeks otherwise. Cilantro bolts quickly in heat; succession-plant every 3–4 weeks or focus on a fall crop.

11. Spinach

- Start indoors: 4–6 weeks before last frost
- Transplant outdoors: 4–6 weeks before last frost (very cold-hardy)
- Container size: 6–8" deep, any width
- Light: Medium. Among the most shade-tolerant vegetables you can grow.
- Notes: Like lettuce, spinach is a cool-season crop. It's one of the first things you can start in late winter and one of the first things you can put outside. Bolt-resistant varieties (like Tyee) are worth the search.

12. Zucchini / Summer Squash

- Start indoors: 2–3 weeks before last frost (no earlier — they grow very fast)
- Transplant outdoors: After last frost
- Container size: 10–15 gallon minimum. This is a big plant. One compact variety per large container.
- Light: High light
- Notes: Choose compact or "patio" varieties like Astia or Bush Baby. Standard zucchini will take over a balcony. Even compact varieties need hand-pollination in balcony environments where pollinators are scarce — use a small paintbrush to transfer pollen from male to female flowers.

Part 2: Container Size Guide

Crop	Minimum Container	Better Choice
Tomatoes (cherry)	5-gallon	7–10 gallon
Tomatoes (full-size)	10-gallon	15-gallon
Peppers	3-gallon	5-gallon
Eggplant	5-gallon	7-gallon
Basil	6" pot	8–10" pot

Lettuce	6" deep window box	8" deep, any width
Kale / Chard	8" deep, 12" wide	12" deep per plant
Cucumbers	5-gallon	7-gallon
Bush beans	8" deep, 12" wide	12" deep, 18" wide
Radishes	6" deep	8" deep
Parsley / Cilantro	6" pot	8-10" pot
Spinach	6" deep	8" deep
Zucchini (compact)	10-gallon	15-gallon

A note on drainage: Every container needs drainage holes. If yours doesn't have them, drill them. Waterlogged roots kill plants faster than anything else in the list.

Part 3: Light Requirements

What "Full Sun" Actually Means Indoors

Outdoor "full sun" means 6–8 hours of direct sunlight. Indoors, even your best south-facing window delivers significantly less light intensity than the same sun outdoors — glass filters UV, and the sun's angle changes throughout the day. A window that gets direct sun for 4 hours isn't the same as being outside in the sun for 4 hours.

This matters because it sets realistic expectations. You can grow vegetables indoors in windows. You'll get better results with supplemental light. And some crops — leafy greens, herbs, cool-season vegetables — genuinely do well in lower-light situations.

Window Growing: What Works, What Doesn't

South-facing window (best): Gets the most sun hours. In spring and summer, this can support tomatoes, peppers, basil, and other high-light crops if you have a full, unobstructed exposure. In winter, it's marginal for fruiting crops.

East-facing window (good): Morning sun, afternoon shade. Excellent for lettuce, spinach, herbs, kale. Not enough for tomatoes or peppers trying to fruit.

West-facing window (decent): Afternoon sun. Similar to east for growing purposes — good for leafy greens and herbs, challenging for fruiting crops.

North-facing window (limited): Ambient light only. Stick to herbs like mint and chives, or sprouts. Don't try to grow tomatoes here.

Practical tip: Get a cheap light meter app on your phone. Hold your phone at plant level and take a reading at noon on a clear day. 2,000–4,000 lux is low light (leafy greens, herbs). 4,000–10,000 lux is medium (most vegetables). 10,000+ lux is high (fruiting crops, basil). Most south-facing windows in North American apartments max out around 8,000–12,000 lux on a clear day, dropping significantly in winter.

Grow Light Setups

You don't need an expensive setup. Here's what works:

Budget option — T5 or T8 fluorescent shop lights: Available at hardware stores for \$20–\$40. Work well for seed starting and cool-season crops. Not ideal for fruiting tomatoes and peppers long-term, but fine for getting seedlings going.

Mid-range — LED grow lights (quantum board style): \$40–\$120 from various brands. Look for full-spectrum LEDs with a color temp around 3000–4000K for growing or 5000–6500K for seed starting. Mars Hydro, AC Infinity, and similar brands offer good value. A 45W–100W fixture covers a standard shelf or small table.

Rules of thumb:

- Keep lights 2–4 inches above seedlings (closer for T5/T8, check manufacturer specs for LEDs)
- Run lights 14–16 hours per day for seed starting
- Use a \$10 outlet timer so you don't have to think about it
- As seedlings grow, raise the light rather than moving the plants

Signs your plants need more light: Stretching toward the window, pale or yellowish leaves, thin stems, slow growth despite correct watering and temperature.

Part 4: Soil Mix Recommendations

Why Regular Garden Soil Doesn't Work

Garden soil — the stuff in your yard or in cheap bags labeled "topsoil" — compacts in containers. It doesn't drain well, roots suffocate, and you get all the pathogens of outdoor soil without any of the natural drainage benefits of being in the ground. Don't use it.

What to Use Instead

For most vegetables and herbs: A peat- or coco coir-based potting mix with added perlite. Look for bags labeled "potting mix" (not "garden soil" or "topsoil"). Brands like Fox Farm Happy Frog, Espoma Organic Potting Mix, or generic quality potting mixes from garden centers work fine.

Improving what you have: Add 20–25% perlite by volume to any standard potting mix. This improves drainage and aeration dramatically, especially important if you're growing in containers that are prone to sitting wet.

DIY mix for the committed:

- 1 part peat moss or coco coir
- 1 part perlite
- 1 part compost (homemade or bagged)
- Optional: a tablespoon of worm castings per gallon for slow-release fertility

For seed starting specifically: Use a finer-textured seed starting mix for germination trays. This stuff has minimal nutrients (seeds don't need nutrients to germinate) and a fine, even texture that makes good seed-to-soil contact easier. Once seedlings have their first true leaves, you can transplant into regular potting mix.

Fertilizing in Containers

Container plants need more regular feeding than ground-planted crops because nutrients wash out with every watering. Start fertilizing once seedlings have their first true leaves:

- Liquid fish/kelp fertilizer (diluted to half strength, every 1–2 weeks): Gentle, organic, works well
- Slow-release granular fertilizer worked into the top inch of soil at transplant: Lower-maintenance option
- Worm castings top-dressed monthly: Gentle, essentially impossible to overapply

Don't fertilize during germination. Don't fertilize stressed plants. When in doubt, give less rather than more — overfertilizing symptoms (burned leaf edges, excessive leafy growth at the expense of fruit) are common in container gardens.

Part 5: Thinning and Transplanting

Thinning: Why It's Necessary and Why It Feels Wrong

When you sow multiple seeds per cell (which you should, as insurance against poor germination), you'll end up with multiple seedlings in the same spot. Thinning means removing the extras so only the strongest seedling remains.

This is hard for new growers to do. It feels wasteful. Do it anyway. Crowded seedlings compete for light, water, and nutrients. They grow weak and get sick easier. One strong plant will always outperform three struggling ones.

How to thin: Use scissors to snip the weakest seedlings at soil level. Don't pull them out — pulling can disturb the roots of the seedling you're keeping. Thin to one seedling per cell as soon as the first true leaves appear (these are the second set of leaves, after the round seed leaves that appear first).

Transplanting Seedlings: Step by Step

- 1. Harden off first.** Before moving seedlings from indoors to outdoors, they need to adjust to outdoor conditions — wind, temperature fluctuation, more intense light. Start by putting them outside in a sheltered spot for 1–2 hours, then bring them back in. Increase outdoor time over 7–10 days until they're spending full days outside. Skip this step and you'll stress or kill plants that took you weeks to grow.
 - 2. Transplant in the evening or on a cloudy day.** This reduces transplant shock by giving plants time to recover before facing full sun.
 - 3. Water the seedling before removing it from its cell.** Moist soil holds together; dry soil crumbles and disturbs roots.
 - 4. Handle by the soil mass or the leaves, never the stem.** Stems are fragile; a pinched stem usually means a dead plant. Leaves can be replaced; stems cannot.
 - 5. Plant at the right depth.** Most vegetables should be planted at the same depth they were growing, with one exception: tomatoes can (and should) be planted deeply, burying the stem up to the lowest set of leaves. Roots will form along the buried stem, making a stronger plant.
 - 6. Water in well after transplanting.** Then let the plant settle for a few days before resuming normal watering schedules.
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Part 6: Troubleshooting

Leggy Seedlings (Tall, Thin, Flopping Over)

What it looks like: Seedlings stretching up with long gaps between leaves, thin stems that can't support themselves.

What's causing it: Not enough light. Almost always.

Fix: Move the light source closer. If using a window, supplement with a grow light. If already using a grow light, lower it (check manufacturer minimums — most LEDs can go to 6–12 inches for seedlings). Once leggy, you can't un-leggy a seedling, but you can stop further stretching and let the plant continue to develop.

Prevention: Adequate light from the moment of germination. Have your light ready before seedlings emerge, not after.

Damping Off (Seedlings Collapse at Soil Line)

What it looks like: Seedlings that were fine yesterday suddenly topple over. Stem looks pinched or rotted right at the soil surface. Often spreads rapidly through a tray.

What's causing it: Fungal disease, typically Pythium or Fusarium. Thrives in cool, wet, poorly ventilated conditions.

Fix: Remove affected seedlings immediately. Improve airflow (a small fan running nearby works well). Let the soil surface dry slightly between waterings. Unfortunately, seedlings that have damped off cannot be saved.

Prevention:

- Don't overwater (the most common contributing factor)
 - Use sterile seed starting mix, not garden soil
 - Run a small fan near seedlings to improve air circulation
 - Water in the morning so surfaces dry during the day
 - Don't reuse last year's seed starting mix without sterilizing it
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Slow Germination

What it looks like: Seeds planted 2+ weeks ago with no sign of germination.

Common causes and fixes:

Soil too cold: Most vegetables germinate best between 65–75°F. In a cool apartment, soil temps near a window or on a floor can be significantly lower. Solution: Use a seedling heat mat. Even a basic \$20 mat makes a significant difference for peppers, eggplant, and tomatoes.

Seeds too deep: Most small seeds should be planted 1/4" deep or less. If you planted at 1/2" or more, they may be struggling to reach the surface. Next time, plant shallower.

Seeds too old: Germination rates decline with age. If you're using seeds from several years ago, especially if stored poorly, germination will be erratic. Do a germination test before sowing: place 10 seeds on a damp paper towel, fold it up, keep it moist and warm for the expected germination time, and count how many sprout. Below 50% germination means you should plant heavily or get fresh seeds.

Soil too dry or too wet: Seeds need consistent moisture to germinate. Cover seed trays with a humidity dome (or plastic wrap) until germination, then remove it immediately when you see sprouts.

Yellow Leaves

Uniform yellowing of lower/older leaves: Usually nitrogen deficiency. Feed with a balanced liquid fertilizer at half-strength.

Yellow leaves with green veins: Usually iron or magnesium deficiency, often caused by pH problems. Check soil pH if possible (inexpensive testers available online). Most vegetables want 6.0–7.0.

Yellowing combined with wet soil: Likely overwatering or poor drainage. Let the soil dry out more between waterings.

Pests: Fungus Gnats

What they are: Tiny flies hovering around your plants and soil. The adults are annoying but largely harmless. The larvae live in soil and can damage roots.

Fix: Let the top inch of soil dry out between waterings — larvae need moisture to survive. Yellow sticky traps catch adults. For serious infestations, a soil drench with *Bacillus thuringiensis israelensis* (BTi, sold as Mosquito Bits or Gnatrol) is organic and effective.

That's the kit. You have everything you need to get started. Your first season is about learning your specific space — how your windows change with the seasons, how your apartment holds temperature, how your plants tell you what they need. Pay attention, take notes, and you'll be a better grower by fall than you are today.

— Seedwarden

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